

Aided with the **folding** and **cut-and-tape** methods, more complex—*multiple- motif design units*—may be achieved. The skeletal or fully figurative units are characterized by order and simplicity through their tile-like positioning resulting in remarkably complex designs.

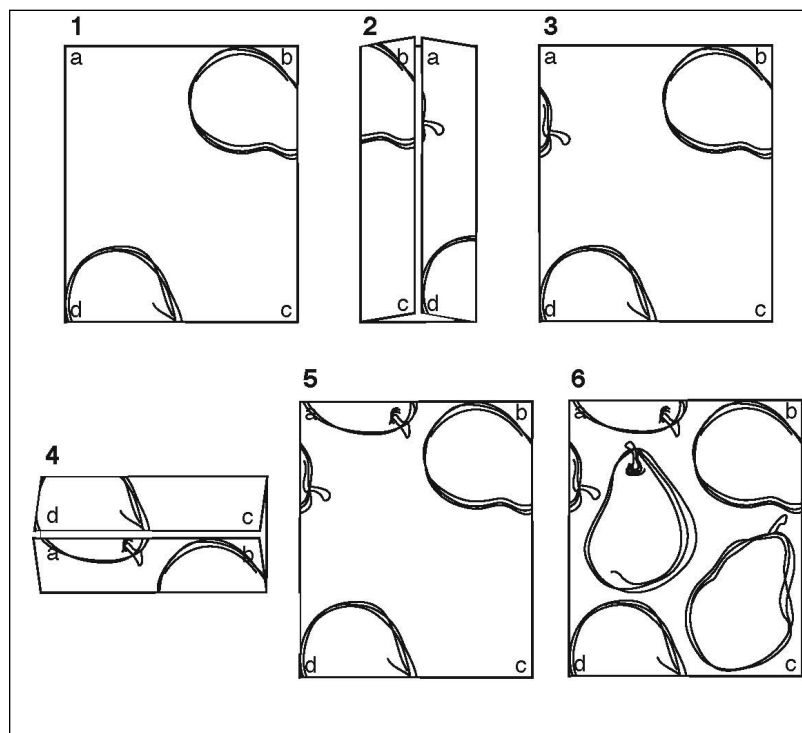
The following methods describe a very straight forward approach to resolving straight-repeat units⁽¹⁾.

Straight repeat units

FOLDING METHOD

The folding method helps create a repeat unit which can then be placed together with its copies in a side-repeat construction.

1. Starting with a sheet of drawing paper, carefully measure and cut out a rectangle or square of a good size (at least 6"x6") to represent the first single repeat. Using a pencil, start to draw your design motifs on the edges (top right, bottom left), or in the centre, if you prefer.
Fold the paper right side out, so that two opposite sides come together without overlapping (sides BC to AD in FIGURE ONE). Draw more of the design over these adjoining edges, completing those started in the top right edge.
2. Open the paper out.
3. Bring the opposite sides together (AB to DC in FIGURE ONE), and again add more of the design or complete the motifs in these matched edges.



5. Unfold the paper, continue to draw motifs until...

6. ...the design is completed. In this example the repeat size is 6.75"x8".

7. Draw a black outline around the repeat unit.

8. You can proceed to photocopy the repeat unit several times (at least 4) and splice the units to make a working repeat. OR, you can scan/photograph it in order to directly clean/tweak it in Photoshop (reposition, re-scale, colour, etc). It may also be re-traced by hand (using tracing paper or a light table) or in Illustrator.

FIGURE ONE

CUT AND TAPE METHOD

A straight repeat may also be established by a method of cutting and taping.

1. On a piece of paper previously measured to the repeat size you wish to work with (a perfect square or rectangle), draw any motifs you like, maybe centred or a bit to the side, but never reaching the edges.
2. Then, draw pencil lines to divide the paper in four even parts and number each quadrant from 1 to 4 in clock wise order.

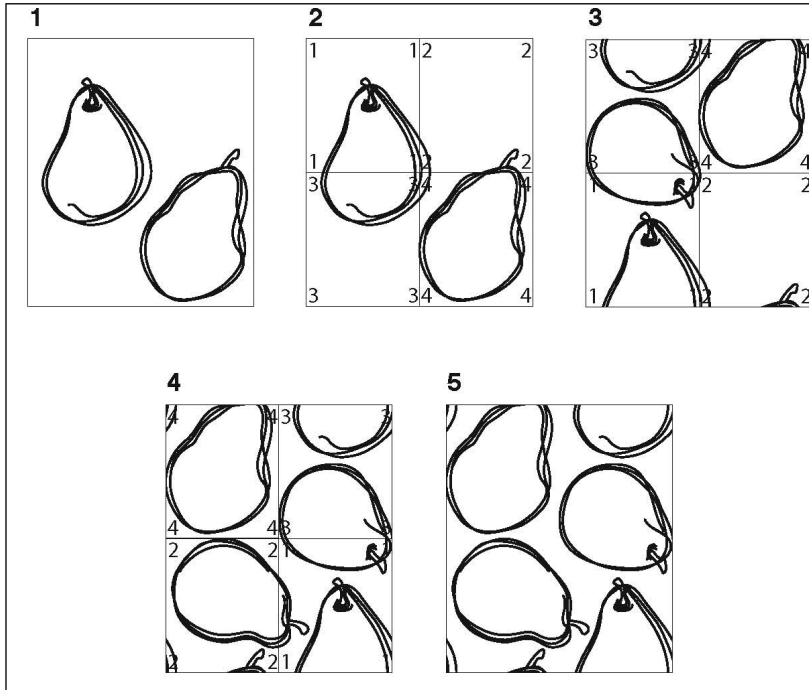


FIGURE TWO

3. Proceed to make a cut on the horizontal, and invert the order of the quadrants, so that 3 and 4 are at the top, and 1 and 2 are at the bottom (FIGURE TWO). Secure the position by taping the back and draw more motifs.

4. Next, make a cut on the vertical, and flip the quadrants once more, so that now 4 and 3 are at the top, and 2 and 1 are at the bottom (FIGURE TWO). Secure the position by taping the back and draw more motifs.

5. The new repeat unit is done.

As mentioned before, you can proceed to photocopy the repeat unit several times (at least 4) and splice the units to make a working repeat. OR, you can scan/photograph it in order to directly clean/tweak it in Photoshop (reposition, re-scale, colour, etc). It may also be re-traced by hand (using tracing paper or a light table) or in Illustrator.

(1) V. Ann Waterman, "Surface Pattern Design", Hastings House Publishers, 1984.